

## Task 1: Probe focus identification (Qwen2-Audio-7B)

Logistic regression probe (PCA-50 + L2) trained on mean-pooled encoder hidden states. LOO cross-validation on Speaker 0 f-files ( $N = 93$ ); held-out test on ns-files. Chance = 50%.

| Layer     | LOO acc      | 95% CI              | SP0-ns        | SP1-ns | SP2-ns |
|-----------|--------------|---------------------|---------------|--------|--------|
| 0         | 69.9%        | [59.9, 78.3]        | 100.0%        | 50.0%  | 54.2%  |
| 1         | 80.6%        | [71.5, 87.4]        | 100.0%        | 54.2%  | 62.5%  |
| 2         | 76.3%        | [66.8, 83.8]        | 100.0%        | 54.2%  | 66.7%  |
| 3         | 79.6%        | [70.3, 86.5]        | 100.0%        | 54.2%  | 75.0%  |
| 4         | 86.0%        | [77.5, 91.6]        | 100.0%        | 54.2%  | 54.2%  |
| 5         | 87.1%        | [78.8, 92.5]        | 100.0%        | 54.2%  | 62.5%  |
| 6         | 81.7%        | [72.7, 88.3]        | 100.0%        | 58.3%  | 54.2%  |
| 7         | 84.9%        | [76.3, 90.8]        | 100.0%        | 58.3%  | 58.3%  |
| 8         | 83.9%        | [75.1, 90.0]        | 100.0%        | 58.3%  | 50.0%  |
| 9         | 75.3%        | [65.6, 82.9]        | 100.0%        | 50.0%  | 54.2%  |
| 10        | 79.6%        | [70.3, 86.5]        | 100.0%        | 50.0%  | 54.2%  |
| 11        | 82.8%        | [73.9, 89.1]        | 100.0%        | 58.3%  | 54.2%  |
| 12        | 74.2%        | [64.5, 82.0]        | 100.0%        | 50.0%  | 54.2%  |
| 13        | 79.6%        | [70.3, 86.5]        | 100.0%        | 50.0%  | 50.0%  |
| 14        | 83.9%        | [75.1, 90.0]        | 100.0%        | 54.2%  | 70.8%  |
| 15        | 81.7%        | [72.7, 88.3]        | 100.0%        | 45.8%  | 58.3%  |
| 16        | 84.9%        | [76.3, 90.8]        | 100.0%        | 58.3%  | 45.8%  |
| 17        | 84.9%        | [76.3, 90.8]        | 100.0%        | 50.0%  | 54.2%  |
| 18        | 83.9%        | [75.1, 90.0]        | 100.0%        | 50.0%  | 54.2%  |
| 19        | 83.9%        | [75.1, 90.0]        | 100.0%        | 50.0%  | 58.3%  |
| 20        | 88.2%        | [80.0, 93.3]        | 100.0%        | 58.3%  | 62.5%  |
| <b>21</b> | <b>89.2%</b> | <b>[81.3, 94.1]</b> | <b>100.0%</b> | 50.0%  | 62.5%  |
| 22        | 83.9%        | [75.1, 90.0]        | 100.0%        | 50.0%  | 50.0%  |
| 23        | 88.2%        | [80.0, 93.3]        | 100.0%        | 62.5%  | 50.0%  |
| 24        | 82.8%        | [73.9, 89.1]        | 100.0%        | 66.7%  | 66.7%  |
| 25        | 84.9%        | [76.3, 90.8]        | 100.0%        | 58.3%  | 54.2%  |
| 26        | 87.1%        | [78.8, 92.5]        | 100.0%        | 62.5%  | 58.3%  |
| 27        | 86.0%        | [77.5, 91.6]        | 100.0%        | 50.0%  | 54.2%  |
| 28        | 80.6%        | [71.5, 87.4]        | 100.0%        | 50.0%  | 50.0%  |
| 29        | 82.8%        | [73.9, 89.1]        | 100.0%        | 50.0%  | 50.0%  |
| 30        | 80.6%        | [71.5, 87.4]        | 100.0%        | 50.0%  | 50.0%  |
| 31        | 72.0%        | [62.2, 80.1]        | 100.0%        | 50.0%  | 54.2%  |
| 32        | 66.7%        | [56.6, 75.4]        | 100.0%        | 45.8%  | 50.0%  |

Table 1: Layer-by-layer focus probe accuracy. Layer 0 = conv embedding, layers 1–32 = Whisper transformer. Peak at layer 21. SP0-ns  $N = 24$ , SP1-ns  $N = 24$ , SP2-ns  $N = 24$ . Chance = 50%.

#### Task 4: Test inference (Qwen2-Audio-7B, output token)

Zero-shot inference classification (A/B/C) using the same prompt as `audioInput.py`. Model output token scored against gold label. Chance = 33.3%.

| Set          | Correct | $N$ | Accuracy | 95% CI       |
|--------------|---------|-----|----------|--------------|
| SP0 f-files  | 27      | 93  | 29.0%    | [20.8, 38.9] |
| SP0 ns-files | 10      | 24  | 41.7%    | [24.5, 61.2] |
| SP1 ns-files | 6       | 24  | 25.0%    | [12.0, 44.9] |
| SP2 ns-files | 7       | 24  | 29.2%    | [14.9, 49.2] |

Table 2: Qwen2-Audio-7B zero-shot inference accuracy (output token, matched prompt). SP0 f-files excludes 7 items with missing audio clips. All CIs overlap chance (33.3%), confirming at-chance performance.